

Entity Analysis Report

Project

E-Commerce Order Management Database System

Entity Identification and Attributes

Entities:

Customer, Category, Product, Orders, OrderItem, Payment, Shipping, Admin.

Customer(CustomerID PK, Name NOT NULL, Email UNIQUE NOT NULL, Phone UNIQUE, Address NOT NULL, Password NOT NULL)

Category(CategoryID PK, CategoryName UNIQUE NOT NULL)

Product(ProductID PK, ProductName NOT NULL, Price CHECK>0, Stock DEFAULT 0 CHECK>=0, CategoryID FK)

Orders(OrderID PK, OrderDate DEFAULT CURRENT_DATE, Status DEFAULT 'Pending', CustomerID FK)

OrderItem(OrderItemID PK, OrderID FK, ProductID FK, Quantity CHECK>0, UnitPrice CHECK>0)

Payment(PaymentID PK, OrderID FK UNIQUE, Amount CHECK>0, PaymentMethod NOT NULL, PaymentStatus DEFAULT 'Pending')

Shipping(ShippingID PK, OrderID FK UNIQUE, Address NOT NULL, ShippingStatus DEFAULT 'Processing')

Admin(AdminID PK, Name NOT NULL, Email UNIQUE NOT NULL, Password NOT NULL)

Foreign Keys:

Product.CategoryID -> Category.CategoryID

Orders.CustomerID -> Customer.CustomerID

OrderItem.OrderID -> Orders.OrderID

OrderItem.ProductID -> Product.ProductID

Payment.OrderID -> Orders.OrderID

Shipping.OrderID -> Orders.OrderID